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Optimizing Weak Reference Processing in the JVM Z Garbage Collector

A Performance Analysis and Optimisation of the ZGC

Reference Processing Pipeline



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1. Introduction

Java is one of the most widely deployed programming languages in enterprise and server-side computing, where applications routinely manage heaps of tens to hundreds of gigabytes. In these environments, the performance of the Java Virtual Machine's (JVM) garbage collector (GC) directly affects application latency and throughput. The Z Garbage Collector (ZGC) addresses this by performing nearly all garbage collection work concurrently with the application, achieving sub-millisecond pause times regardless of heap size.

However, one aspect of ZGC that has received comparatively little optimisation attention is its processing of *weak references*. Java's `WeakReference` class allows applications to hold references that do not prevent the referent from being collected, and it is widely used in caching libraries, canonical mappings, and listener registrations. When the garbage collector determines that a weakly referenced object is no longer strongly reachable, it must clear the reference and if the reference was registered with a `ReferenceQueue`, enqueue it for the application to process.

The current ZGC reference processing pipeline treats all weak references uniformly: they are linked together in an intrusive linked list, processed, and then handed to a Java-level daemon thread (the `ReferenceHandler`) for enqueueing. This design introduces overhead that is disproportionate for a specific subset of weak references: those that are not registered with a `ReferenceQueue`.

This thesis investigates whether targeted modifications to ZGC's reference processing pipeline can reduce this overhead. Three orthogonal optimisation approaches are explored: skipping the enqueue path for references without queues, replacing the intrusive linked list with a cache-friendly dynamic array, and reducing per-reference instruction cost through general code optimisations.

1.1 Motivation

Weak references are handled very specifically in JVM garbage collection. Unlike regular references, which are handled implicitly by the mark phase, weak references require explicit processing by the collector. Each discovered weak reference must be individually evaluated, its referent field cleared if the referent is unreachable, and the reference potentially enqueued. This per-reference processing cost scales linearly with the number of weak references, making it a bottleneck in applications that allocate millions of them.

The problem is amplified by a mismatch between the API design and its use cases. Java's `WeakReference` API provides an optional `ReferenceQueue` parameter, and the garbage collector's pipeline is designed around the assumption that cleared references will always be enqueued. This assumption is still made even if the optional `ReferenceQueue` parameter was omitted. The result is a pipeline that performs unnecessary work for weak references that lack a `ReferenceQueue`.

ZGC's concurrent architecture makes this inefficiency particularly relevant. Because ZGC processes references concurrently with the application, any reduction in per-reference overhead translates directly into less time spent in the reference processing phase. This results in less CPU time consumed by GC threads, ultimately improving application performance.

1.2 Problem Statement

The current ZGC reference processing pipeline does not distinguish between weak references with and without a `ReferenceQueue` during its phases. All cleared references are threaded onto a pending list, transferred to the `ReferenceHandler` thread, and iterated regardless of whether they will actually be enqueued. Additionally, the intrusive linked-list data structure used for discovered references exhibits poor cache locality.

This thesis seeks to determine whether these inefficiencies can be addressed through modifications to ZGC's reference processor, and to quantify the resulting performance impact.

1.3 Research Questions

The main research questions that this thesis aims to address are:

1. What design changes and optimisations can be made to the processing of weak references without a `ReferenceQueue` in ZGC to improve the GC's performance?
2. How do these design changes and optimisations impact the overall performance and memory footprint of ZGC?

1.4 Stretch Goals

In addition to the primary research questions, the following stretch goals are defined to explore the broader implications of this work:

1. Does the introduction of a weak field annotation in Java provide a more efficient way to handle weak references without a `ReferenceQueue` compared to the current `WeakReference` object-based implementation?

2. Can the optimisations developed for ZGC be applied to other garbage collectors in the JVM, such as the G1 garbage collector, and if so, what performance improvements can be observed?

1.5 Scope and Delimitations

This thesis focuses on the reference processing pipeline within ZGC, specifically targeting weak references without a `ReferenceQueue`. Although the optimisation approaches are described in general terms and the stretch goals consider a wider scope, the primary implementation only targets ZGC's reference processor. Furthermore, some of the optimisations, such as general code optimisations, may have broader applicability beyond just weak references without a `ReferenceQueue`. However, these broader applications fall outside the scope of this thesis and are not evaluated.

The correctness of the optimisations is verified through the standard OpenJDK test suite and manual inspection; formal verification is outside the scope of this work. Performance measurements are conducted on a single hardware platform under controlled conditions so therefore may not generalize to all environments.

2. Background

The three optimisation approaches introduced in Chapter 1 rely on several key concepts and design decisions in the Java Virtual Machine’s (JVM) garbage collection architecture and specifically the Z Garbage Collector (ZGC) and its reference processing pipeline. This chapter provides the necessary background on these topics to understand the motivation and implementation of the optimisations.

2.1 Garbage Collection in the JVM

The JVM provides automatic memory management through garbage collection (GC). Rather than requiring the programmer to explicitly allocate and free memory as in languages such as C or C++, the JVM tracks which objects are reachable from the application’s root set and reclaims the memory of objects that are no longer reachable. The root set consists of all data that is directly accessible to the application, such as local variables on the stack, static fields, and active threads.

The JVM ships with several garbage collector implementations, each designed with different trade-offs between throughput, latency, and memory footprint [1]. The collectors are:

- **Serial GC** A single-threaded collector that performs all garbage collection work using one thread, avoiding inter-thread communication overhead. It is best suited to single-processor systems and small heaps.
- **Parallel GC** A throughput-oriented generational collector that uses multiple GC threads to speed up collection on multiprocessor systems. It is intended for medium to large data sets.
- **G1 GC (Garbage-First)** A mostly concurrent, region-based collector designed to scale from small to large multiprocessor machines. It targets predictable pause times with high probability while maintaining high throughput and is the default collector on most platforms.
- **ZGC (Z Garbage Collector)** A scalable, low-latency collector that performs expensive GC work concurrently with application threads. It targets pause times of only a few milliseconds (largely independent of heap size) and supports heap sizes from 8 MB to 16 TB.

This thesis focuses on ZGC, as its low-latency design and concurrent reference processing pipeline can particularly benefit from optimisations that reduce the amount of work done during reference processing.

2.1.1 Garbage Collection Phases

Garbage collection generally occurs multiple times (cycles) during the lifetime of a Java application. Each cycle consists of several distinct phases, each with specific responsibilities. The exact phases and their order can vary between collectors, but a common structure is as follows:

1. **Mark Phase** The collector identifies all live objects by traversing from the root set and marking all reachable objects. Dead objects are those that are not marked. This phase may be performed concurrently with the application or during a stop-the-world pause, depending on the collector design.
2. **Reference Processing Phase** After marking completes, the collector processes *Reference* objects discovered during the mark phase. This phase determines which references have dead referents and prepares them for enqueueing. Like marking, this phase may be concurrent or done during a pause.
3. **Sweep/Reclamation Phase** The collector reclaims memory occupied by dead objects. Some collectors use a sweep algorithm (scanning the heap and freeing unmarked objects) whilst others use evacuation (copying live objects to new memory regions).
4. **Compaction/Relocation Phase** Some collectors compact the heap to eliminate fragmentation, moving live objects to consolidate free space. Others relocate objects incrementally as part of the collection process.

The order and concurrency model of these phases varies significantly between collectors. Serial and Parallel GC typically perform all phases in stop-the-world pauses. G1 performs marking and sweeping mostly concurrently with short pauses for synchronisation. ZGC performs marking, reference processing, and relocation concurrently, using barriers to maintain consistency with concurrent application threads.

2.1.2 Generational Collection

All JVM garbage collectors (Serial, Parallel, G1, and Z) employ *generational collection*, based on the *weak generational hypothesis*: most objects die young and the ones that don't tend to live for a long time.[2] To employ this the heap is divided into at least two generations:

- The **young generation**, where newly allocated objects reside. Young generation collections are frequent but fast, since most young objects are short-lived.
- The **old generation**, where objects that have survived multiple young collections are promoted to. Old generation collections are less frequent but more expensive since live objects are more plentiful here.

ZGC adopted generational collection starting with JDK 21. In generational ZGC, the young and old generations can be collected independently and there-

fore concurrently. This is relevant to reference processing because references in different generations are handled during different collection cycles.

2.2 Java Reference Types

In addition to ordinary (strong) references, Java provides *reference objects* in the `java.lang.ref` package that allow applications to reference objects without preventing their garbage collection. These reference types are:

1. **SoftReference** The referent field is cleared at the discretion of the GC, typically when memory is low. Commonly used for memory-sensitive caches.
2. **WeakReference** The referent field is cleared as soon as it becomes weakly reachable (i.e., no strong or soft references exist to it). Widely used in caching frameworks, canonical mappings, and listener registrations.
3. **FinalReference** An internal JVM type (not part of the public API) used to implement the `finalize()` mechanism. Objects with finalisers are wrapped in a `FinalReference` so the GC can schedule finalisation before reclaiming the object. This scheduling is done via a `ReferenceQueue` which means that `FinalReferences` always have an associated `ReferenceQueue`.
4. **PhantomReference** The referent is never accessible through the reference; `get()` always returns `null`. Used for post-mortem cleanup actions, often to avoid the pitfalls of finalisation. Phantom references must be registered with a `ReferenceQueue` to be useful.

All four types extend the abstract class `Reference<T>` (in the `java.lang.ref` package), which defines the key fields of the reference object and the API for retrieving the referent, clearing the referent field, and checking if it is enqueued.

2.2.1 The Reference Object Lifecycle

Each `Reference` object maintains several fields that track its state:

- `referent` field The referenced object. The GC may clear this field (set it to `null`) when the referent becomes eligible for collection according to the reference's type.
- `queue` field The `ReferenceQueue` this reference is registered with. If the reference was not created with a queue, this field points to the sentinel object `ReferenceQueue.NULL_QUEUE`.
- `next` field Used as a link pointer when the reference is enqueued in a `ReferenceQueue`.
- `discovered` field A field with dual purpose: during GC marking, it serves as a link in the garbage collector's internal *discovered list*; after

processing, it is reused as a link in the VM's *pending list* that feeds the `ReferenceHandler` thread. Due to its first use, the GC can detect whether a reference has already been discovered by checking if this field is `null`.

A reference object transitions through the following states during its lifetime:

1. **Active** The reference has been created and the referent is still reachable (or not yet processed).
2. **Pending** The GC has determined the referent is eligible for clearing and has added the reference to the VM's pending list. The `ReferenceHandler` thread will process it.
3. **Enqueued** The `ReferenceHandler` thread has moved the reference into its associated `ReferenceQueue`, where the application can retrieve it via `poll()` or `remove()`.
4. **Inactive** The reference has been dequeued by the application, or was created without a queue and has been cleared. It is no longer tracked by the GC or queuing infrastructure.

An important observation is that references created *without* a `ReferenceQueue` can transition directly from Active to Inactive, bypassing the Pending and Enqueued states entirely. However, as discussed in Section 2.4, the default implementation does not take advantage of this shortcut.

2.2.2 WeakReference and ReferenceQueue

A `WeakReference` is the most commonly used non-strong reference type in Java applications. It has two constructors, see Listing 1:

```
1 // Without a queue - queue field set to NULL_QUEUE
2 public WeakReference(T referent) {
3     super(referent);
4 }
5
6 // With a queue
7 public WeakReference(T referent, ReferenceQueue<? super
8     T> q) {
9     super(referent, q);
10 }
```

Listing 1: `WeakReference` constructors

When a `WeakReference` is created with a `ReferenceQueue`, the application can be notified when the referent is collected. When created *without* a queue, the reference serves only as a conditional pointer, the application can call `get()` to check whether the referent is still alive, but receives no notification upon clearing.

The `ReferenceQueue` is implemented as a synchronized singly-linked list using the `next` field of the `Reference` objects. It provides `poll()` for non-blocking retrieval and `remove()` for blocking retrieval with an optional timeout. This is the mechanism by which applications can react to the collection of referents, for example to clean up associated resources.

A key insight for this thesis is that many real-world uses of `WeakReference` does not require a `ReferenceQueue`. Caching frameworks, `WeakHashMap`, and similar data structures typically create weak references without queues, relying instead on periodic polling or lazy cleanup. This means the garbage collector expends effort preparing these references for enqueueing (adding them to the pending list) even though the `ReferenceHandler` thread will immediately skip them.

2.3 The Z Garbage Collector

ZGC is a concurrent, region-based, compacting garbage collector included in the JDK since JDK 11 (experimental) and production-ready since JDK 15. Starting with JDK 21, ZGC supports generational collection, which is now the default mode. ZGC's primary design goals are:

- Sub-millisecond pause times that do not increase with heap size.
- Support for heap sizes from 8 MB to 16 TB.
- Concurrent marking, relocation, and reference processing.

ZGC achieves these goals through several key techniques:

Coloured Pointers.

ZGC embeds metadata in the unused bits of 64-bit object pointers. These *colour bits* encode information about the pointer's current state relative to the GC's phase, for example, whether the pointer has been remapped after relocation, or whether it points to a young or old generation object. This allows the GC to reason about pointer states without loading the object header.

Barriers.

ZGC uses barriers at every reference load or write. The barrier checks the colour bits of the loaded pointer and, if necessary, applies a fixup (e.g., remapping after relocation). This is the mechanism that allows ZGC to perform most of its work concurrently with the application.

Concurrent Phases.

A ZGC collection cycle interleaves short stop-the-world pauses (for root scanning and synchronisation) with concurrent phases (for marking, reference processing, and relocation). The concurrent phases run alongside application threads, using barriers to maintain consistency.

Multiple GC Worker Threads.

ZGC uses multiple GC worker threads to parallelize GC work. Each worker thread maintains its own data structures in many cases to avoid contention.

2.4 Reference Processing Pipeline in ZGC

Reference processing is the mechanism by which the garbage collector handles `Reference` objects discovered during marking. The pipeline has three stages: *discovery*, *processing*, and *enqueueing*. In ZGC, the first two stages run concurrently with the application during the marking phase, while the enqueue stage involves handing off references to the Java-level `ReferenceHandler` thread.

2.4.1 Discovery

During the concurrent marking phase, when ZGC encounters a `Reference` object (an instance of `SoftReference`, `WeakReference`, `FinalReference`, or `PhantomReference`), it evaluates whether the reference should be *discovered*, i.e. added to a list for later processing.

The discovery decision depends on several factors:

1. **Is the reference active?** Only active references are eligible. For non-`FinalReference` types, an inactive reference has a null referent field. For `FinalReference`, inactivity is indicated by a non-null `next` field.
2. **Is the referent still strongly reachable?** If the referent is already marked as strongly live, the reference does not need processing. The garbage collector checks this via its liveness information.
3. **Soft reference policy.** For `SoftReference` objects, the collector applies a policy (based on available memory and the reference's last access time) to decide whether to keep the referent alive or clear the referent field.
4. **Generational considerations.** In ZGC, only references residing in the old generation are discovered; young-generation references are always treated as strong. This is because reference processing introduces additional delay and uncertainty to the duration of the young-generation collection. So by treating young references as strong, ZGC can complete young collections more predictably which decreases the risk of the application running out of memory during a long young collection.

Once a reference is selected for discovery, it is recorded on a *discovered list*. In ZGC each GC worker thread maintains its own discovered-list head, so discovery is implemented as a cheap, non-atomic prepend to a per-worker list. This avoids CAS contention found in shared processors where multiple threads update the same discovered head.

2.4.2 Processing

After marking completes, the discovered lists are processed. For each discovered reference, the collector determines the final fate of the referent:

1. If the referent field has become `null` (cleared by the application concurrently), the reference is dropped from the list.
2. If the referent is now strongly reachable (e.g. it was marked as alive after the reference's discovery), the reference is dropped.
3. If the referent is still unreachable according to the reference's strength, the referent field is **cleared** (set to `null`), and the reference is prepared for enqueueing.

In ZGC, clearing involves setting the `referent` field to a coloured null pointer via compare-and-swap (CAS) to ensure thread safety since application threads may be concurrently accessing the reference.

After clearing, the reference is linked onto the thread-local pending list. This involves writing the reference's address into the `discovered` field of the previously processed reference. Once the thread has processed all references in its discovered list the thread-local list gets atomically prepended to the ZGC's internal pending list.

Generational interactions are important during processing: when a `Reference` object is old but its referent is young, the referent is treated as strongly reachable and the referent field must not be cleared. Processing must therefore verify generation membership and, if appropriate, ensure young-generation liveness is preserved (by marking the young referent).

2.4.3 Enqueue

After a reference's referent field has been cleared, the reference must be made available to the application if it was created with a `ReferenceQueue`. This happens through the *pending list* mechanism which feeds the `ReferenceHandler` thread. The handoff process is as follows:

1. The GC has built an internal pending list by linking cleared references together via their `discovered` field.
2. Under the heap lock, the GC atomically prepends its internal pending list onto the global VM pending list.
3. The `ReferenceHandler` daemon thread is notified.
4. The `ReferenceHandler` thread wakes up, atomically retrieves the pending list, and iterates over it. For each reference, it checks whether `queue != NULL_QUEUE`. If the reference has a real queue, it calls `queue.enqueue(this)`; otherwise it simply advances to the next reference.

This handoff reveals an inefficiency: references created without a `ReferenceQueue` are still linked into the pending list, transferred to the `ReferenceHandler` thread, and iterated over only to be skipped. For work-

loads where the majority of weak references lack queues, this results in a considerable amount of extra work. This behaviour has been reported in an OpenJDK enhancement issue[\[3\]](#). Avoiding enqueue-path work for references with `NULL_QUEUE` is one of the primary optimisation target explored in this thesis.

3. Method

In order to answer the research questions posed in Section 1.3 three orthogonal optimisation approaches were implemented separately for the processing of weak references without a `ReferenceQueue` in the Z garbage collector (ZGC). These separate implementations were then combined in all possible combinations and had their performance evaluated using microbenchmarks. The three approaches were, as introduced in Chapter 1, the following:

- Skipping the enqueue path for weak references without a `ReferenceQueue`
- Replacing the intrusive linked list with a dynamic array for the discovered list
- General Code Optimisations

The rationale, implementation details, and performance testing methodology of these approaches are described in the following sections. The evaluation of the approaches is described in Chapter 4.

3.1 Skipping Enqueue Path

As described in Section 2.4.2 ZGC enqueues discovered references to the internal pending list regardless of whether they have a `ReferenceQueue` or not. This is inefficient for weak references without a `ReferenceQueue` as discussed in Section 2.4.3. Because of this, the reference processing pipeline was changed so that the enqueue path can be skipped for weak references without a `ReferenceQueue`. The intended effect of this optimisation is to both reduce the amount of work done by the GC threads during reference processing and to reduce the amount of work done by the `ReferenceHandler` thread when iterating over the pending list.

3.1.1 Implementation of Null-queue Fast Path

In order to skip enqueueing weak references without a `ReferenceQueue` the `ReferenceProcessor` class was modified to check if the queue field of a weak reference being discovered is equal to the sentinel value `ReferenceQueue.NULL_QUEUE` before adding it to the discovered list. If the queue field is equal to `ReferenceQueue.NULL_QUEUE` (the weak reference wasn't created with a queue) it is added to a separate discovered list for weak

references without a `ReferenceQueue` instead of being added to the regular discovered list. This separate discovered list is then processed independently by the GC threads and none of its weak references are enqueued to the pending list for the `ReferenceHandler` thread. Each GC thread maintains its own separate discovered list for weak references without a `ReferenceQueue` to avoid contention between threads.

The reason for separating the discovered lists instead of simply skipping enqueueing for weak references without a `ReferenceQueue` is that two unique reference processing functions can then be used for the two lists. One function that includes enqueueing and one that doesn't. This allows for more efficient processing of the weak references without a `ReferenceQueue` as it avoids unnecessary checks for whether the weak reference has a queue during processing.

3.2 Replacing Intrusive Linked List with Dynamic Array

When processing the discovered references, ZGC iterates over the discovered list. This list is, as described in Section 2.4.1, implemented as an intrusive linked list of the discovered references. This iteration over a linked list is inefficient since it has poor cache locality and cannot exploit instruction-level parallelism between iterations. Because of this, the intrusive linked list was replaced with a dynamic array for the discovered list for weak references without a `ReferenceQueue`. The intended effect of this optimisation is to reduce the amount of time spent loading the next reference to process and thereby allow for more efficient processing of the discovered references through better cache locality and better exploitation of instruction-level parallelism.

3.2.1 Implementation of Dynamic Array for Discovered References

The implemented dynamic array, called `ZAddressArray`, stores discovered references in a contiguous layout rather than linking them via the `discovered` field. The array is allocated on the C heap and grows using a power-of-two strategy (with a minimum capacity of 8), using `memcpy` for bulk copying during expansion rather than element-by-element loops.

Each GC worker thread maintains its own `ZAddressArray` instance to avoid contention. During discovery, when a reference should be added to a discovery list its data is appended to the worker's array instead of being appended to a linked list. The reference's `discovered` field is still set (to a self-referencing pointer) to mark the reference as discovered, preventing duplicate discovery.

During processing, the array is iterated sequentially using index-based access. Because the elements are stored contiguously in memory, this iteration benefits from hardware prefetching and improved cache line utilisation compared to following pointers scattered across the heap.

When using both dynamic arrays and separate discovered lists, only the discovered list for weak references without a `ReferenceQueue` is backed by a `ZAddressArray`, while the regular discovered list continues to use the linked list.

3.3 General Code Optimisations

The standard reference processing path in ZGC uses general-purpose functions designed to handle all reference types uniformly. This generality introduces per-reference overhead that is unnecessary when processing only weak references without a `ReferenceQueue`, where the reference type is always `REF_WEAK` and enqueueing is never required.

The standard function for determining whether a reference's referent should be cleared, performs several operations per reference:

1. A load barrier to read the referent from the reference object.
2. A virtual call to determine the reference type (soft, weak, final, or phantom).
3. A compare-and-swap (CAS) operation via a ZGC barrier to atomically set the referent field to a coloured null pointer.

For weak references without a `ReferenceQueue`, each of these operations can be simplified or eliminated:

- The reference type is known to be `REF_WEAK` at the call site, eliminating the need for the virtual call to determine the type.
- The only concurrent modification to the referent field that can occur is from the application threads, which can only change it from a live referent to null. This allows the CAS operation to be replaced with a simple store of a coloured null pointer without needing to check for concurrent updates.
- When combined with the dynamic array optimisation, the referent field address and its value can be pre-fetched during discovery and stored in the array entry, avoiding redundant memory loads during processing. This is because during the reference processing phase, the only field that can be modified by another thread is the referent field, and it can only be modified from a live referent to null. This means that all values can be safely pre-fetched as long as the referent field isn't set to a non-null value without any safeguards during processing.

3.3.1 Implementation of General Code Optimisations

The implementation introduces a new function for determining whether a queue-less weak reference's referent should be cleared. This function takes a data struct (either from the dynamic array or constructed on the fly from the linked list) containing the pre-fetched field addresses and values.

The new function checks whether the referent is strongly live via metadata in ZGC's heap management. If the referent is not strongly live and it resides in the old generation, its referent field is cleared using a direct pointer store rather than a CAS. This is okay since, as mentioned previously, the only possible new value is null so the worst-case scenario is that null is overwritten with null. If the referent is still live (either strongly marked or implicitly by residing in the young generation), the referent field is recoloured with the correct pointer metadata bits, this recolouring however has to occur with a CAS so that a null value isn't overwritten with a non-null value. Additionally, if the referent is young and the young generation is currently in its mark phase, the referent is marked to preserve cross-generational liveness.

The discovered field is also cleared using a direct store to the field address rather than through an atomic access function since no other thread modifies this field after discovery.

3.4 Build Configurations

The three optimisation approaches are controlled by compile-time preprocessor flags: `ZUseGenCodeOptimisations` for the general code optimisations, `ZUseSeperateDiscoveredLists` for the separate discovered list, and `ZUseDynamicArray` for the dynamic array. Each flag can be independently enabled (1) or disabled (0) via compiler flags passed during the JDK build.

Eight build configurations were created to evaluate all combinations of the three optimisation flags, as shown in Table 3.1. Each configuration produces a separate JDK build with its own binary, allowing direct comparison under identical benchmark conditions.

Table 3.1. *Build configurations and their enabled optimisation flags*

Configuration	Gen. Opts	Sep. Lists	Dyn. Array
none	—	—	—
dyn_only	—	—	✓
sep_only	—	✓	—
sep_dyn	—	✓	✓
gen_opt_only	✓	—	—
gen_opt_dyn	✓	—	✓
gen_opt_sep	✓	✓	—
all	✓	✓	✓

3.5 Performance Evaluation

To evaluate the performance impact of the optimisation approaches, a microbenchmark was developed that exercises the weak reference processing pipeline under controlled conditions. The benchmark and its execution environment are described below.

3.5.1 Benchmark Design

The benchmark allocates approximately 7.3 million objects, each containing a randomly sized byte array payload (between 509 and 1097 bytes). Each object is held by both a strong reference and a `WeakReference`. Of the weak references, 5% are created with a `ReferenceQueue` and 95% without, reflecting a workload dominated by queue-less weak references. The allocation order of both strong and weak references is randomised using a Fisher-Yates shuffle to avoid artificial locality patterns.

After allocation, the benchmark proceeds through five rounds of clearing. In each round, approximately 20% of the strong references are nulled in a randomised order, making the corresponding objects eligible for garbage collection. Between rounds, the benchmark sleeps to allow GC cycles to discover and process the weak references whose referent now is unreachable. After each clearing round, any remaining strong references are touched (written to) to ensure they remain live and are not optimised away.

The benchmark runs three iterations with a five-second cooldown between them to allow the JVM to stabilise. Each iteration performs the full allocation, clearing, and GC cycle sequence independently.

3.5.2 Execution Environment

To ensure reproducible measurements, the benchmark is executed under controlled conditions using a dedicated runner script. The script applies the following isolation measures:

- **CPU pinning:** The Java process is pinned to a fixed set of CPU cores using `taskset`. The monitoring and scripting processes run on separate cores to avoid interference.
- **CPU frequency control:** The CPU governor is set to performance mode at the base clock frequency. Both the minimum and maximum scaling frequencies are locked to the base frequency to prevent frequency scaling from introducing variance.
- **Process priority:** The Java process is run with elevated priority (`nice -n -5`) to reduce scheduling interference from other system processes.
- **Cache clearing:** Filesystem caches are dropped between runs by writing to `/proc/sys/vm/drop_caches`.

- **Cooldown period:** A configurable cooldown period (default 10 seconds) is inserted between variant runs to allow the system to return to a stable state.

The JVM is configured with a fixed 8 GB heap (`-Xms8g -Xmx8g`), ZGC enabled (`-XX:+UseZGC`), and a forced major collection interval of 0.5 seconds (`-XX:ZCollectionIntervalMajor=0.5`) to ensure frequent GC activity during the benchmark. Tenuring thresholds are set to 1 (`-XX:InitialTenuringThreshold=1 -XX:MaxTenuringThreshold=1`) to promote objects to the old generation quickly, ensuring that the reference processor discovers them. GC statistics and reference processing logs are enabled via `-Xlog:gc+stats,gc+ref`.

3.5.3 Measurement Methodology

The benchmark is run for 100 outer iterations. Within each outer iteration, all eight variants are run in sequence with environment preparation between each. This interleaved ordering mitigates systematic drift in system conditions across the measurement campaign.

During each run, a memory monitoring script samples the process's resident set size (RSS), GC reserved and committed memory (via JVM Native Memory Tracking), and heap usage at high frequency. The current benchmark phase and iteration are tracked from the benchmark's log output to enable phase-aligned analysis.

4. Results

5. Discussion

6. Conclusion

References

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